L11.5 – Train/Test split and Accuracy

Data Mining
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Thanks to Dr Alejandro Arbelaez
And to Flaticon.com

Looking for the perfect classifier

Supervised learning

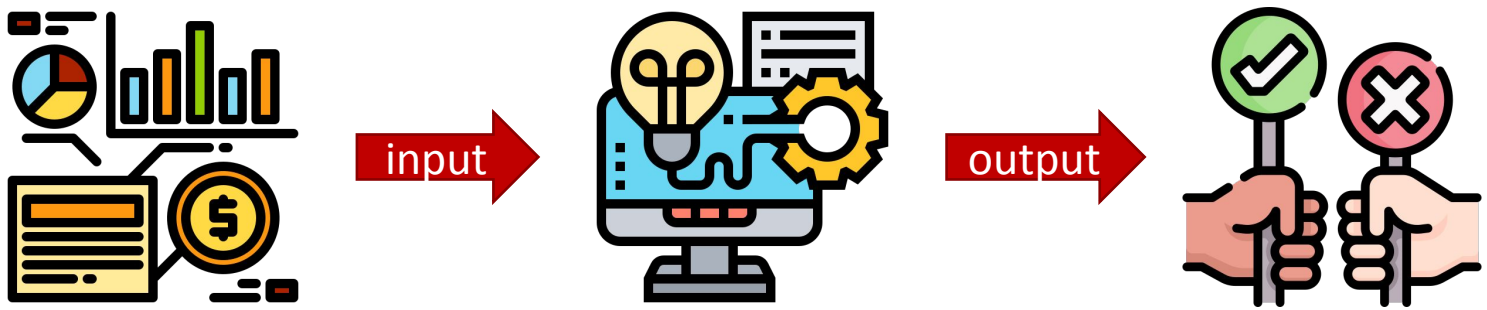
In supervised learning the dataset the dataset contains labels / desired output.

Data

Label

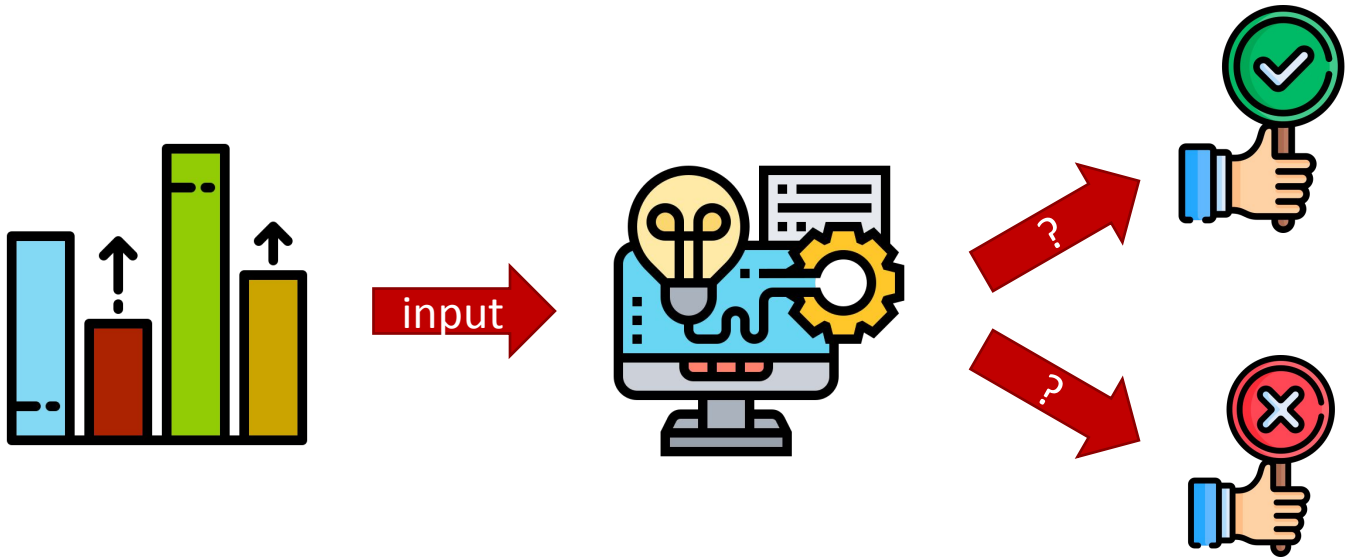
Supervised learning

The machine learning model learns how to infer the desired output based on the input.



Supervised learning

The model then receives previously unseen data and try to infer the label with as much accuracy as possible



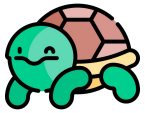
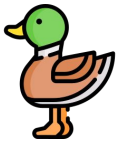
Classification

- Given a collection of records (**training set**)
  - Each record contains a set of attributes, one of the attributes is the class
- Find a **model** for class attribute as a function of the values of other attributes
- Goal: **previously unseen** records should be assigned a class as accurately as possible
  - A **test set** is used to determine the accuracy of the model. Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to validate it

Classification Accuracy

- Most used and simple way to evaluate the performance of a classifier.

$$Accuracy = \frac{\#\ istances\ classified\ correctly}{\#\ instances}$$

Attribute	Label
	Fox
	Panda
	Flamingo
	Duck
	Turtle
	Turtle
	Duck

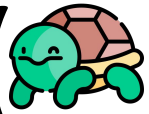
f( ) = Fox

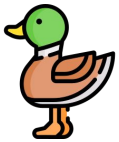
f( ) = Panda

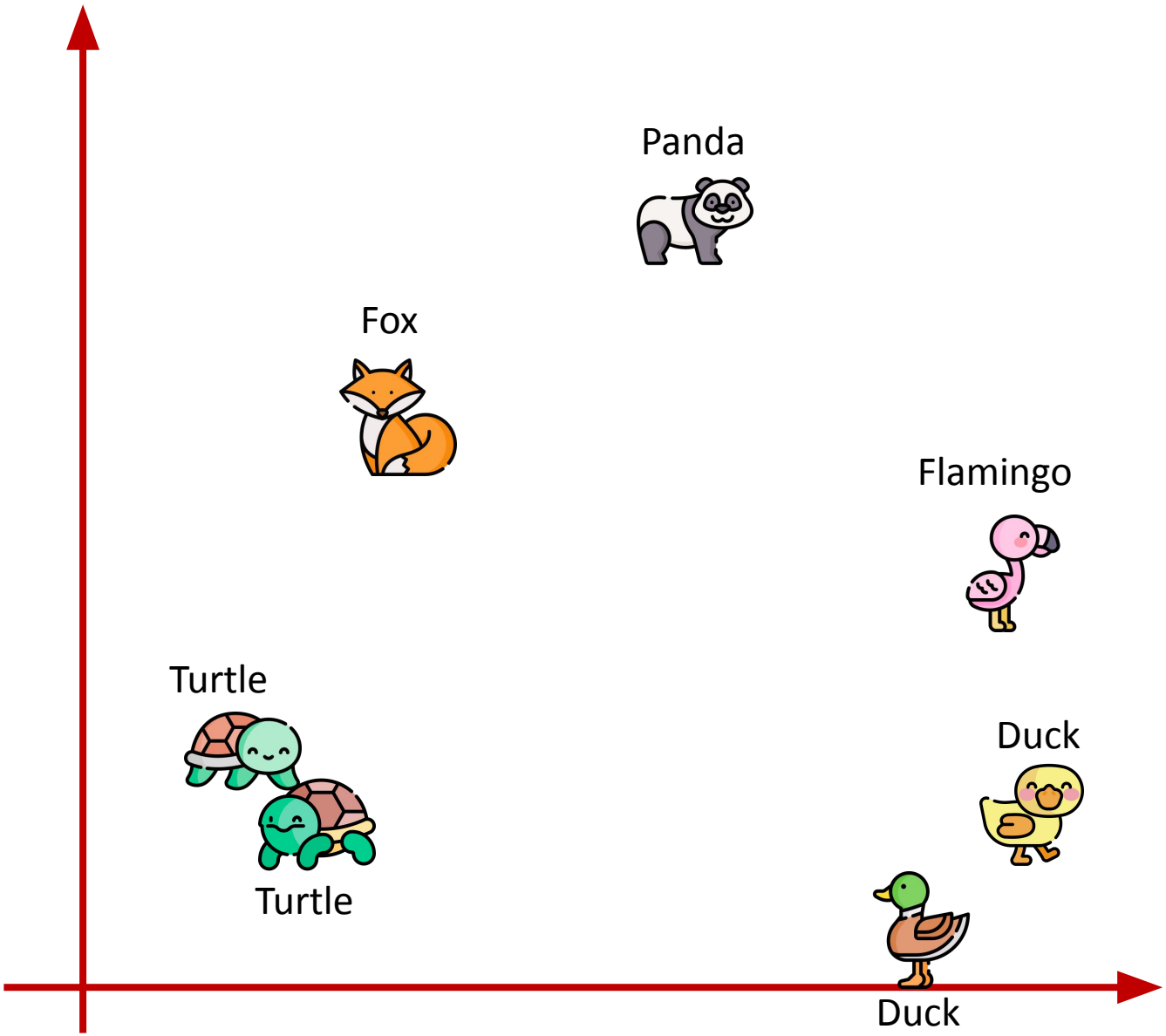
f( ) = Flamingo

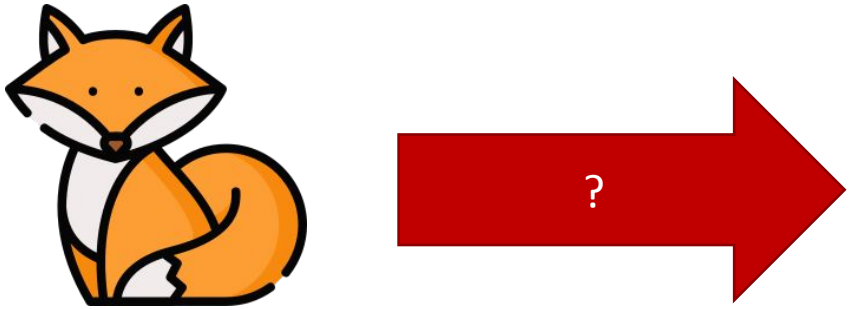
f( ) = Duck

f( ) = Turtle

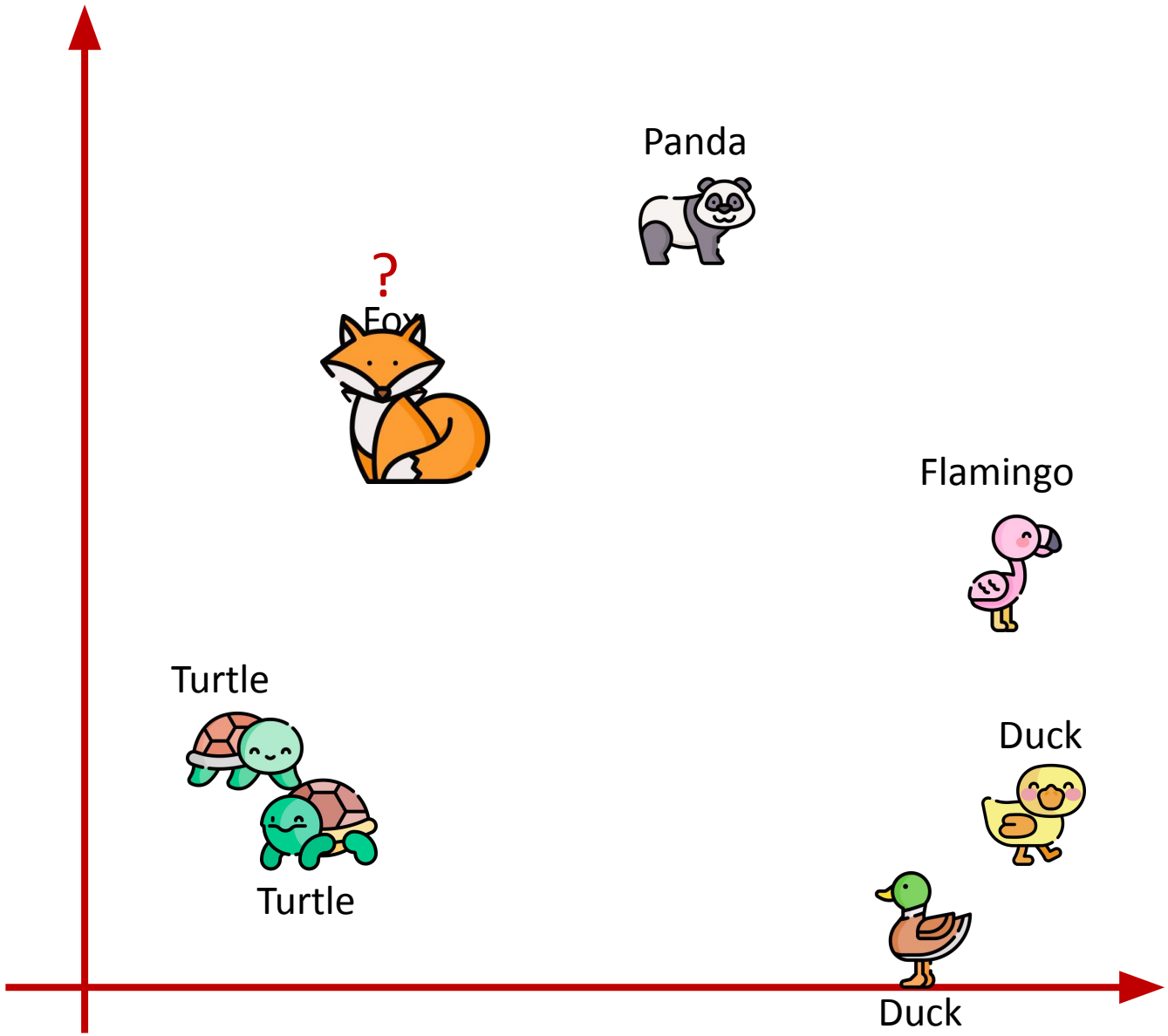
f() = Turtle

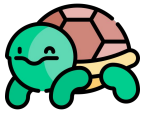
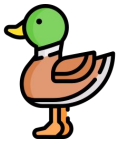
f() = Duck


Panda
Fox
Flamingo
Turtle
Duck
Turtle
Duck



f( ) =


Panda
?
Fox
Flamingo
Turtle
Duck
Turtle
Duck

Attribute	Label
	Fox
	Panda
	Flamingo
	Duck
	Turtle
	Turtle
	Duck

Data Leakage

- The goal is to develop a model that makes accurate predictions on new data, unseen during training.

- The performances should always be evaluated on unseen data.

https://machinelearningmastery.com/data-leakage-machine-learning/

Data Leakage

- Two main causes:
  - You test your model on the same data that it is trained on
  - Your data already contains the answer you are looking for

Data Leakage

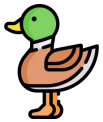
- It is a serious problem for at least 3 reasons:
  - **It is a problem if you are running a machine learning competition**. Top models will use the leaky data rather than be good general model of the underlying problem.
  - **It is a problem when you are a company providing your data**. Reversing an anonymization and obfuscation can result in a privacy breach that you did not expect.
  - **It is a problem when you are developing your own predictive models**. You may be creating overly optimistic models that are practically useless and cannot be used in production.

As machine learning practitioners, we are primarily concerned with this last case.

Test-Train split

- The instances in which we train the model must be different from the ones we are testing it on.

Train Set

Attribute	Label
	Fox
	Panda
	Flamingo
	Duck
	Turtle
	Turtle
	Duck

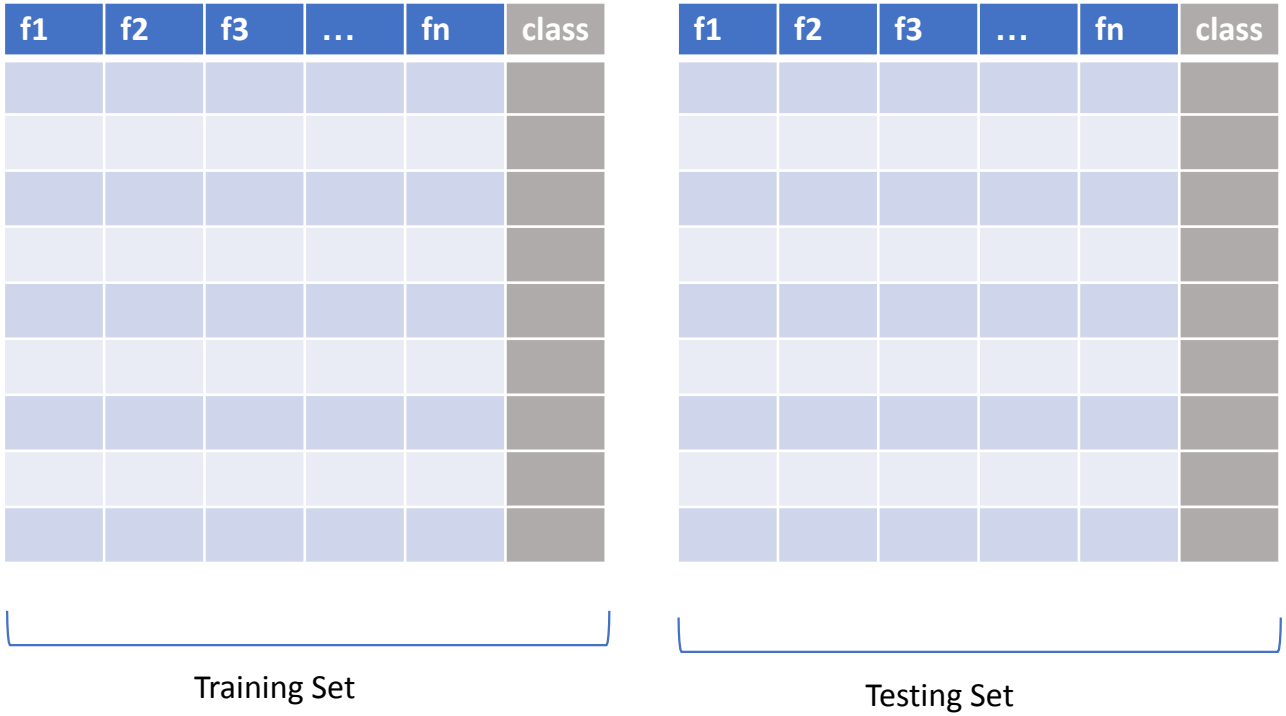
Test Set

Attribute	Label
	Cow
	Panda
	Tucan
	Flamingo

Holdout Set Configuration

- Keeping part of your dataset for test only.

Holdout Set Configuration



Pro and Cons

- The advantage of holdout is:
  - The test error is independent of the training set.
- The disadvantages of this method are:
  - Results can vary quite a lot across different runs.
    - Informally, you might get lucky — or unlucky
  - I.e. in any one split, the data used for training or testing might not be representative.
  - We are training on only a subset of the available dataset, perhaps as little as 50% of it.
    - From so little data, we may learn a worse model and so our error measurement may be pessimistic.
- In practice, we only use the holdout method when we have a very large dataset.
  - The size of the dataset mitigates the above problems.

Question?

- In the k-NN, how do we select the best k?

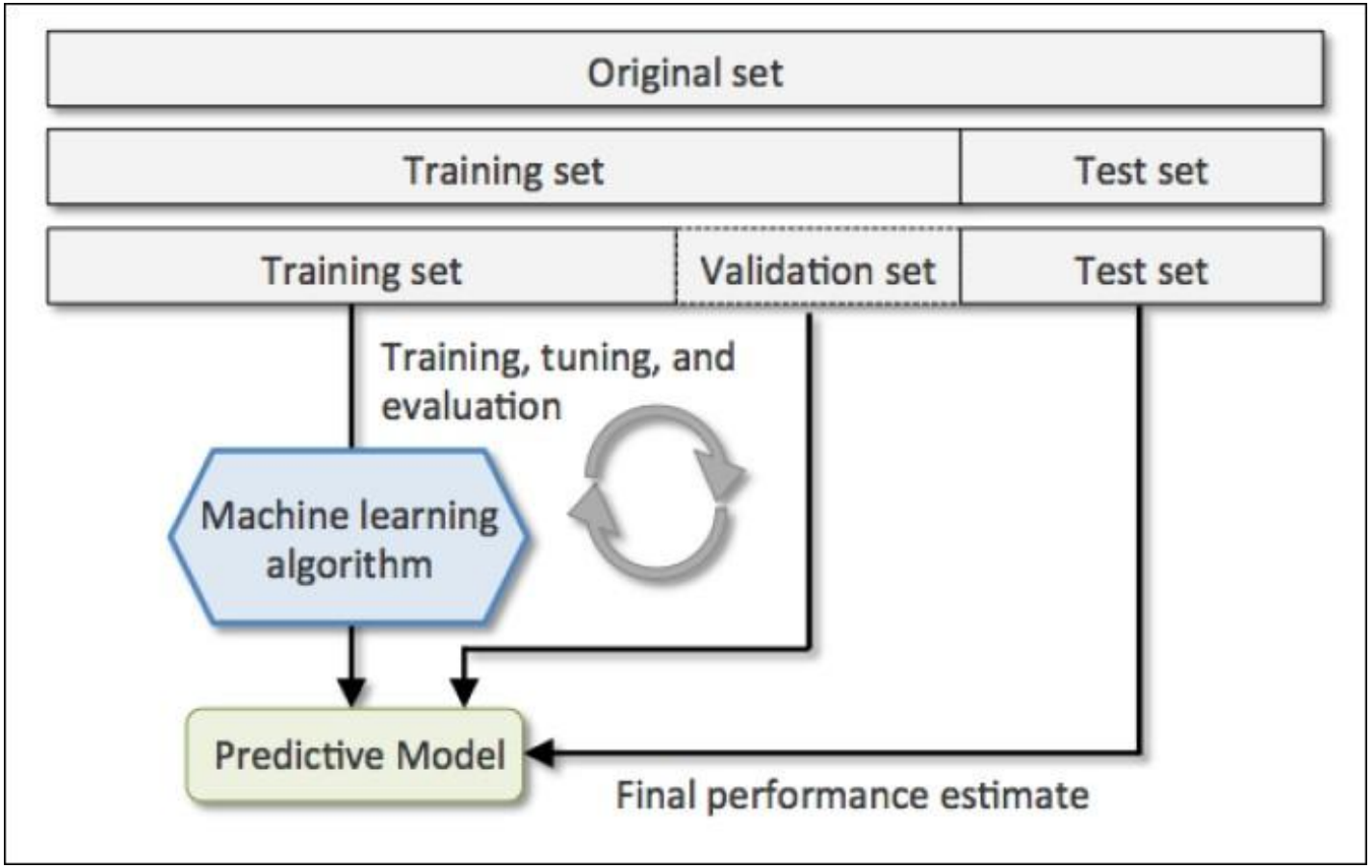
Overfitting

- As shown in the previous slide we are currently using a holdout set configuration. That is, we have a training dataset and we have a test dataset.
- However, if we reuse the same test dataset over and over again during model selection, the model will just **overfit** on the training data using the hyper-parameters.
- This mean that even if we obtain really strong accuracy from model selection using a holdout set then when we deploy the model it won't reach an accuracy anywhere close to this value.
- Unfortunately, despite this very significant issue, many people still use the test set for model selection, which is **not good machine learning practice**.

Holdout Cross-Validation

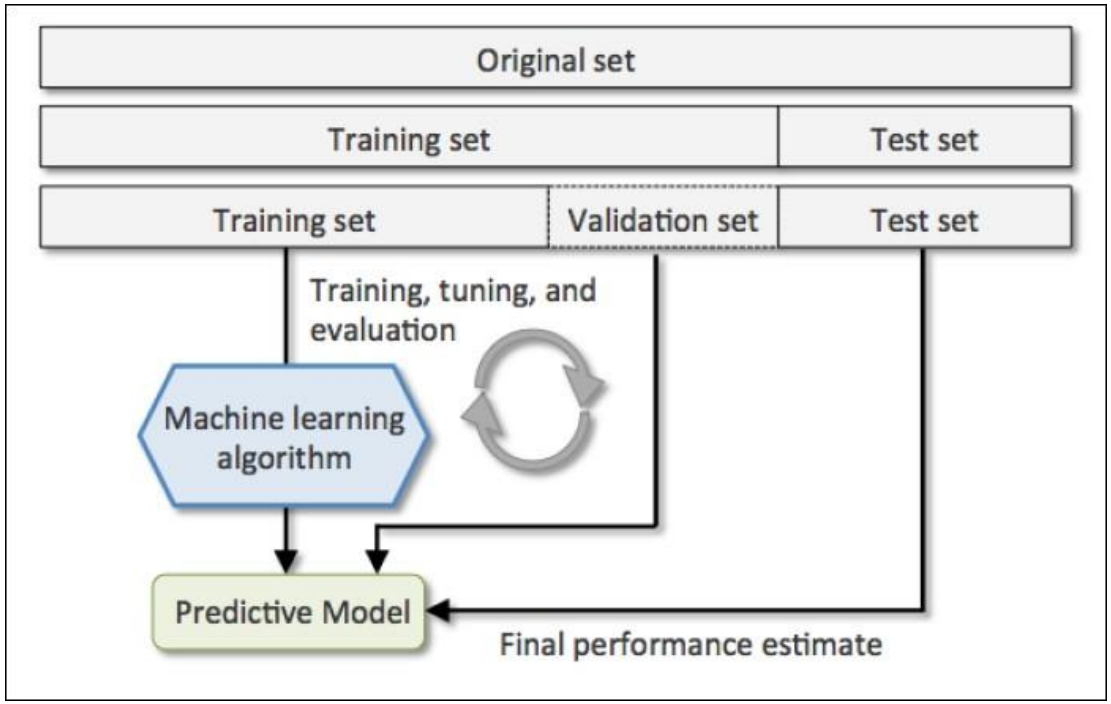
- A better way of using the holdout method for model selection is to separate the data into three parts: a **training** set, a **validation** set, and a **test** set.
- The training set is used to fit the different models, and the performance on the validation set is then used for the model selection (see next slide).
- The advantage of having a test set that the model hasn't seen before during the training and model selection steps is that we can obtain a less biased estimate of its ability to generalize to new data.
- This method is referred to as **holdout cross validation**.

Holdout Cross-Validation



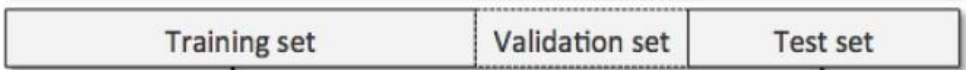
Holdout Cross-Validation

Use a validation set to repeatedly evaluate the performance of the model after training using different parameter values. Once we are satisfied with the tuning of parameter values, we estimate the models' generalization error on the test dataset. Any drawbacks?



The drawback of the holdout cross-validation

- A disadvantage of the holdout cross validation method is that the performance estimate is sensitive to how we partition the training set into the **training** and **validation** subsets.



- In other words the accuracy estimate will vary for different samples of the data.

- A more robust technique for performance estimation, is k-fold cross-validation, where we repeat the holdout method k times on k subsets of the training data

Cross Validation

- Widely used alternative to single Training Set + Validation Set
- Multiple evaluations using different portions of data for training and validating/testing
- k-fold Cross Validation
  - k = number of folds (integer)

k-Fold Cross Validation

- Available data is divided into equal-sized folds (partitions)
- k separate evaluation experiments:
  - Training set: k-1 folds
  - Test set: 1 fold
- Repeat k-1 more times, each using a different fold for the test set

10-fold Cross Validation

10-fold Cross Validation

1/10th of training data

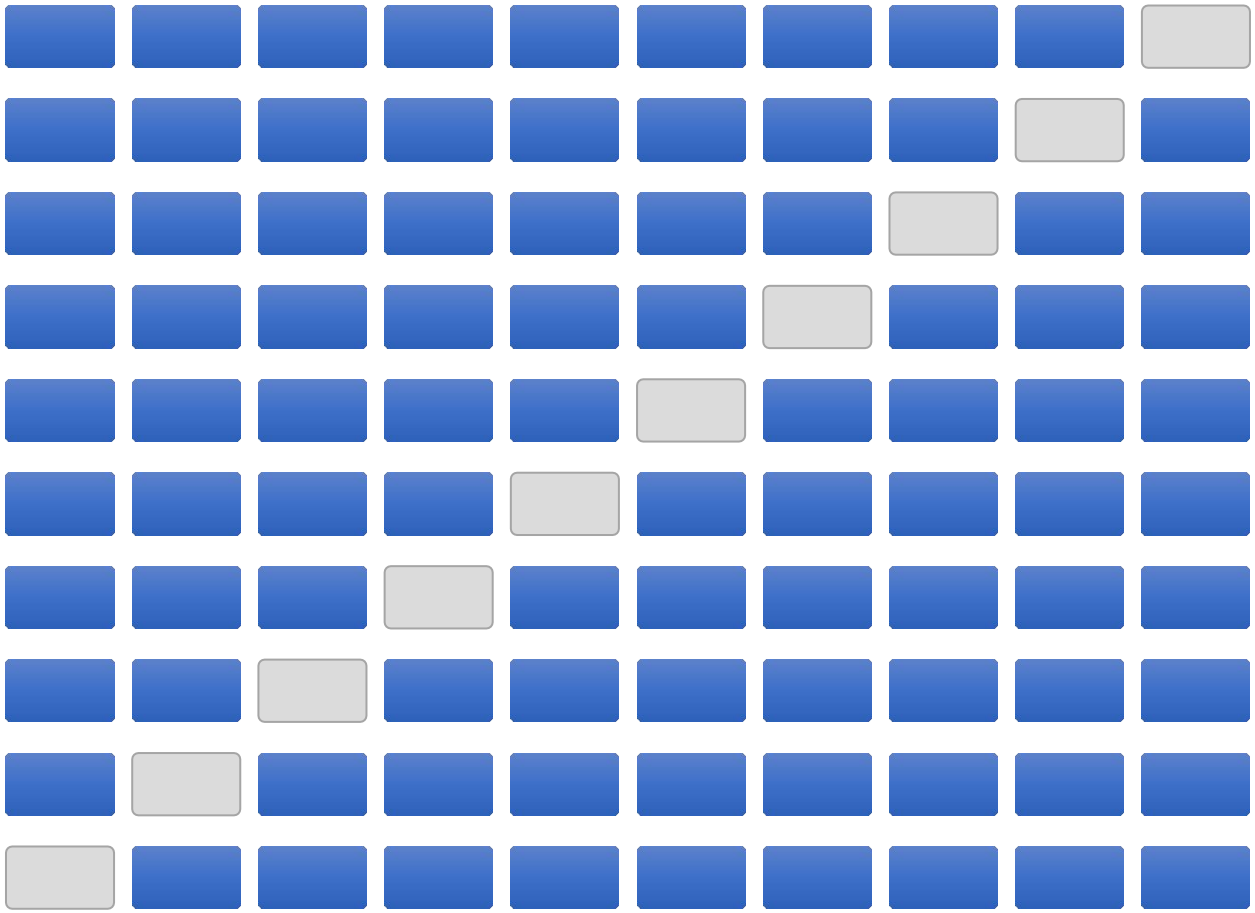
X 9 = 90%

used for training

10% used for testing

- Repeat *k* times
- Average results
- Each instance will be used <u>once</u> in testing

10-fold Cross Validation



1/10th of training data

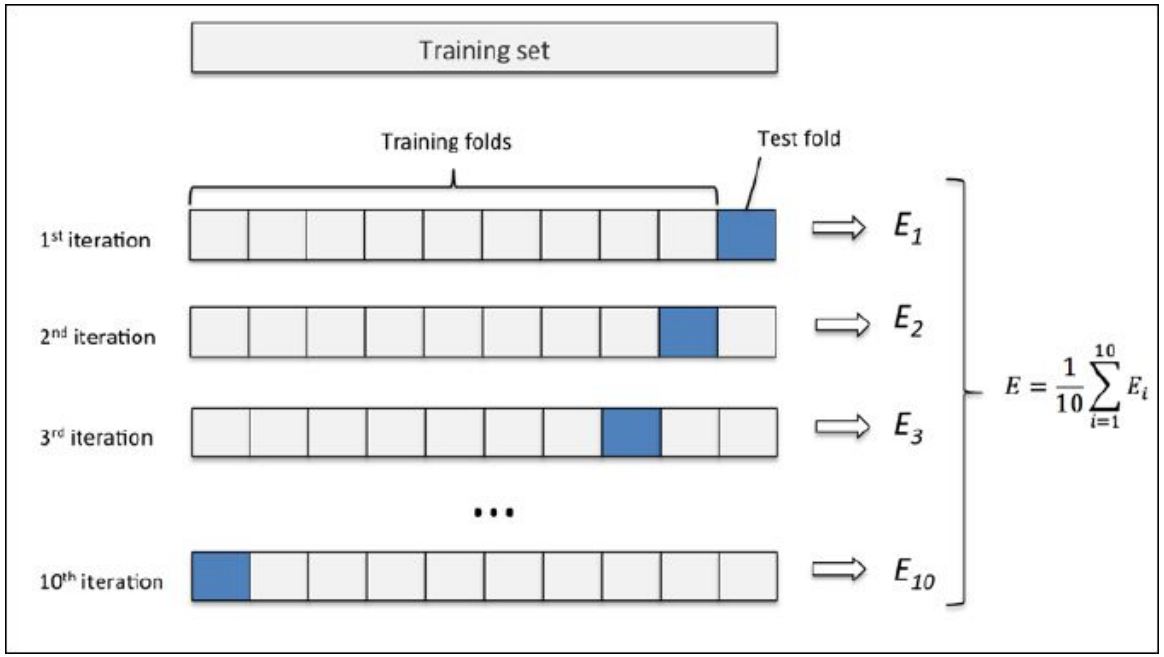
X 9 = 90% used for training

10% used for testing

- Repeat *k* times
- Average results
- Each instance will be used <u>once</u> in testing

K Fold Cross Validation

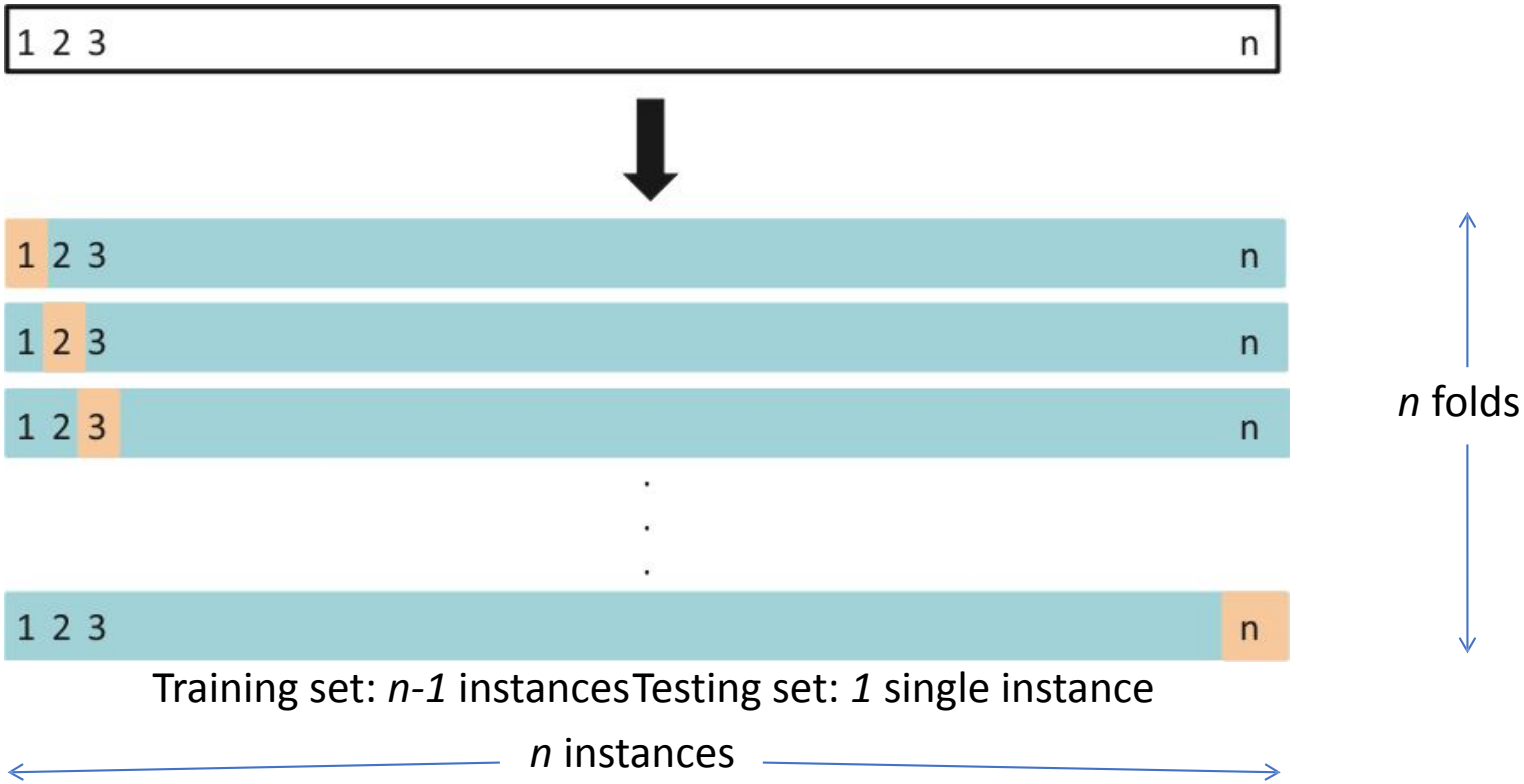
- K-Fold cross validation is also useful for situations where **data is limited**.



Cross-Validation

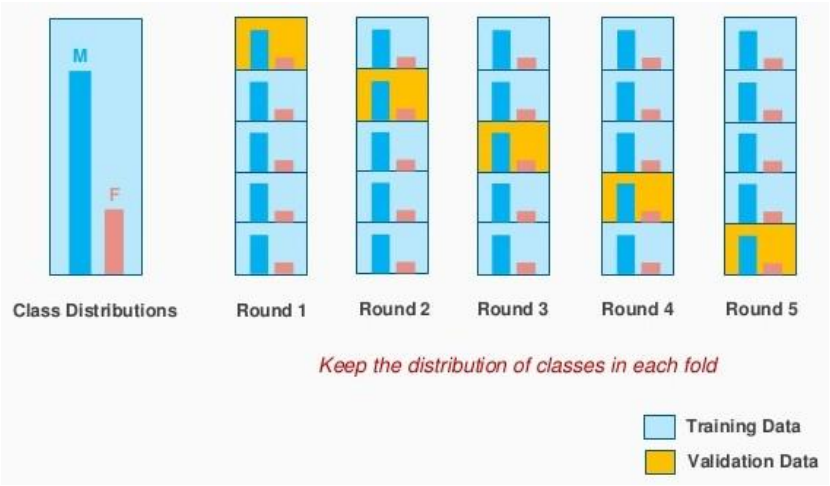
- More computationally expensive
- **k-fold Cross-Validation**
  - k = number of folds (integer)
  - k = 10 is common
- **Leave-One-Out Cross-Validation**
  - Extreme: k = n, where there are n observations in training + validation set
  - Useful when the amount of data available is too small to have a big enough training sets in a k-fold cross-validation.
  - Significantly more computationally expensive

Leave-One-Out Cross-Validation



Stratified Cross Fold Validation

- A slight improvement over the standard k-fold cross-validation approach is stratified k-fold cross- validation, which can yield better bias and variance estimates, especially in cases of unequal class proportions.
- In stratified cross-validation, the **class proportions are preserved in each fold** to ensure that each fold is representative of the class proportions in the training dataset.



m-runs of *k*-fold cross validation

- Another common approach is to run the k-fold cross validation over multiple shuffles of the original dataset.
- Between different shuffles the accuracy can vary considerably (up to 9% in some cases)
  - Visentin, A., Nardotto, A., & O'Sullivan, B. (2019). *Predicting judicial decisions: a statistically rigorous approach and a new ensemble classifier.* In 2019 IEEE 31st International Conference on Tools with Artificial Intelligence (ICTAI)
- Doing multiple shuffles makes the results more stable and reliable.

Evaluation Metrics

Performance Evaluation Metrics

- Basic classification or misclassification accuracy hides quite a lot of detail.

- For example, an ML algorithm might be doing well are predicting one class but may be poor are predicting another class. This type of behaviour can often be masked when simply looking at classification accuracy.

- There are a range of different metrics that can be used to provide a more meaningful evaluation accuracy. Before we delve into these we will first introduce the notion of a confusion matrix.

Confusion Matrix

Confusion matrix contains information about the actual and predicted classifications of a classification algorithm

- **True positives (TP)** - Number of instances of Category A that were correctly classified as Category A.
- **False Positives (FP)** - Number of instances of Category B that were incorrectly classified as Category A.
- **False Negative (FN)** - Number of instances of Category A that were incorrectly classified as Category B.
- **True Negative (TN)** - Number of instances of Category B that were correctly classified as Category B.

		Predicted	
		Category A	Category B
Actual	Category A	*w*	*x*
	Category B	*y*	*z*

Confusion Matrix

Confusion matrix contains information about the actual and predicted classifications of a classification algorithm

- **True positives (TP)** - w
- **False Positives (FP)** - y
- **False Negative (FN)** - x
- **True Negative (TN)** - z

		Predicted	
		Category A	Category B
Actual	Category A	*w* **TP**	*x* **FN**
	Category B	*y* **FP**	*z* **TN**

Multiclass Example

		Predicted		
		Cat	Horse	Dog
Actual	Cat	*10*	*0*	*6*
	Horse	*0*	*15*	*1*
	Dog	*5*	*1*	*9*

We can extract quite a lot of information from a confusion matrix. For example if we focus on the horse class we can see that there are 15 true positives, where a horse is correctly predicted to be a horse. There is only one instance of a horse being incorrectly classified, it is misclassified as a dog. Also notice that no Cat is incorrectly classified as a horse and only a single Dog is classified as a horse.

		Predicted		
		Cat	Horse	Dog
Actual	Cat	*10*	*0*	*6*
	Horse	*0*	*15*	*1*
	Dog	*5*	*1*	*9*

Classification Accuracy

-

Predicted

Actual		Class A	Class B
	Class A	True Positives	False Negatives
	Class B	False Positives	True Negatives

- The correct way to measure accuracy is:

$$Accuracy = \frac{True\ Positive + True\ Negative}{True\ Positive + False\ Negative + False\ Positive + True\ Negatives}$$

That's all Folks!
SUPERB wallpapers